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Gaussian type orbital basis sets for the calculation of continuum properties in molecules: The photoionization cross section of H₂

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The now well-established L^2 techniques applied to the description of the electronic continuum of small molecules and generally implemented with suitable oscillating basis sets are here adapted and used in connection with Gaussian type orbital (GTO) functions. As a test, a K -matrix approach associated with extended sets of GTOs is employed to compute with satisfactory accuracy partial wave phase shifts and both the integral and differential photoionization cross sections of the hydrogen molecule. It is apparently possible to greatly extend the range of applicability of the usual L^2 techniques beyond the limited set of molecular systems amenable to the one center expansion approximation.

I. INTRODUCTION

The calculation of transition matrix elements involving the electronic continuum can be performed by resorting to several techniques. All of them are ultimately characterized by either the use of truly continuum orbitals¹⁻⁵ based on the direct calculation of suitable approximated continuum states with variational or other techniques, and which require the numerical solution of a set of differential or integral equations, or by the use of complete projections on L^2 basis sets.⁶⁻⁸ While the former should in principle afford greater accuracy, the latter has some definite advantages. Algorithms describing on the same basis set both discrete and continuum orbitals are amenable to sequences of matrix operations easily and efficiently implemented on modern computers, and calculations can be carried out by exploiting well established codes developed to treat bound state problems.

The reactance K -matrix method (KM)⁸ originally developed for general variational basis sets, was recently adapted to L^2 basis sets and shown to be extremely effective in computing the quantities needed to obtain differential photoionization cross sections.^{9,10}

In our previous applications of the K -matrix technique particular basis functions were devised to determine the partial wave phase shifts^{10,11} by fitting the asymptotic long range behavior of the variational wave functions lying in the continuum with adequately phase-shifted Coulomb waves. These techniques cannot be applied to GTO basis sets owing to their inadequacy in describing Coulomb waves at large distance from the origin, particularly for high photoelectron energies.

On the other hand, by using a special approach, the K -matrix technique was recently successfully employed to compute partial wave phase shifts for the atom of helium using GTO basis functions.¹² Here this approach is extended to molecules and as a first test application the in-

tegral and differential photoionization cross sections of the hydrogen molecule are computed.

II. THEORY

The theory of the K matrix and its applications in the study of the properties of the electronic continuum in both atoms and molecules have been widely described elsewhere.^{10,13} The approach used here extends the method to the case where short-range basis functions, instead of the more adequate Slater type orbital times cosine (STOCOS) functions,^{10,13} are employed in K -matrix calculations and has been described in a previous paper.¹²

The energy variational K -matrix method is essentially a configuration interaction (CI) in the electronic continuum,⁸ carried out at a fixed energy, formally allowing one to expand the variational solution upon continuous basis sets. The Schrödinger equation is turned into a system of integral equations for the elements of the K matrix and then transformed into a linear system by quadrature upon a finite L^2 set.

The formal basis used to expand the continuum states includes some localized functions (to adequately represent autoionizing states and short-range correlation effects) and the relevant partial wave channel (pwc) subspaces. The pwc basis consists of a set of antisymmetrized spin adapted products of the ion states (whose quantum numbers are collectively indicated by j) and partial l, m waves. Let us indicate the generic pwc basis function by $|\Phi_{\beta E}\rangle$, ($\beta \equiv j_{\beta}, l_{\beta}, m_{\beta}$). The sets are built so that the Hamiltonian is diagonal within each pwc subspace

$$\langle \Phi_{\beta' E'} | \hat{\mathcal{H}} | \Phi_{\beta E} \rangle = \delta_{\beta\beta'} \delta(E - E') E. \quad (1)$$

For a photoionization process of a neutral system, the waves behave asymptotically as shifted Coulomb waves. A fundamental requirement which allows a safe representation upon a discrete basis is that the pwc states should

exhibit properties (phase shifts, off-diagonal overlap, and Hamiltonian matrix elements) weakly dependent on the energy indices. To this end, the localized states are generally included in a separate channel.

If n channels are open at a given energy E , trial variational wave functions $|\Psi_E\rangle = \{|\Psi_{\alpha,E}\rangle\}$ which asymptotically behave as standing waves may be defined, so that

$$|\Psi_{\alpha,E}\rangle = |\Phi_{\alpha,E}\rangle + \sum_{\beta} \oint dE' |\Phi_{\beta,E'}\rangle P \frac{1}{E-E'} K_{\beta,E',\alpha,E} \quad (2)$$

with $\oint$ indicating summation over the discrete part and integration over the continuum part of the spectrum and P denoting the principal part of the integral. The expansion coefficients are determined by imposing

$$\langle \Phi_{\beta,E'} | \hat{\mathcal{H}} - E | \Psi_{\alpha,E} \rangle = 0, \quad (3)$$

which leads to the system of coupled equations

$$\begin{aligned} K_{\beta,E',\alpha,E} - \sum_{\beta'} \oint dE'' V_{\beta,E';\beta',E''}(E) P \frac{1}{E-E''} \\ K_{\beta',E'',\alpha,E}(E) = V_{\beta,E';\alpha,E}(E), \\ V_{\beta,E';\beta',E''}(E) = \langle \Phi_{\beta,E'} | \hat{\mathcal{H}} - E | \Phi_{\beta',E''} \rangle - (E' - E) \\ \times \delta_{\beta,\beta'} \delta(E' - E''). \end{aligned} \quad (4)$$

The resulting wave functions satisfy the relationship

$$\langle \Psi_{E'} | \hat{\mathcal{H}} - E'' | \Psi_E \rangle = (E - E'') [1 + \pi^2 K(E)^2] \delta(E - E'), \quad (5)$$

where $K(E)$ is the so-called K matrix on the energy shell

$$K_{\alpha',\alpha}(E) = K_{\alpha',E,\alpha,E}. \quad (6)$$

Two sets of complex orthonormal eigenfunctions may be obtained by

$$|\Psi_E^{(\pm)}\rangle = \{|\Psi_{\alpha,E}^{(\pm)}\rangle\} = \{|\Psi_{\alpha,E}\rangle\} [1 \pm i\pi K(E)]^{-1} \Delta^{(\pm)}(E), \quad (7)$$

where $\Delta^{(\pm)}(E) = \delta_{\alpha',\alpha} e^{\pm i\xi_{\alpha,E}}$ and $\xi_{\alpha,E}$ is the total phase shift of the pwc basis function $|\Phi_{\alpha,E}\rangle$. The states $|\Psi_{\alpha,E}^{(-)}\rangle$ exhibit the correct asymptotic behavior for photoionization processes.

For real energies, the K matrix on the energy shell is a real symmetric matrix related to the scattering matrix.¹⁴

The pwc phase shifts $\xi_{\alpha,E}$ required to compute the differential cross section were previously obtained¹⁰ by comparing the large r behavior of each partial wave with that of a shifted Coulomb wave of the same energy. This procedure was possible since the L^2 basis included very diffuse functions of the STOCOS type, guaranteeing an accurate description of the wave up to large values of r . With GTO bases, this is very difficult to obtain since, especially for higher values of the wave energy, they may satisfactorily account for the photoejected electron wave only in a rather limited region of space. A different procedure should thus be used. One may observe that if the basis used to represent a single pwc, say α , is built as a product of an ion state times a pure Coulomb wave, then the diagonalization of the Hamiltonian produces pwc states whose phase shift is given by the well-known formula of the K -matrix theory¹²

$$\xi_{\alpha}(E) = \xi_{\alpha,0}(E) - \tan^{-1}[\pi K_{\alpha,\alpha}(E)], \quad (8)$$

where $\xi_{\alpha,0}$ is the phase shift of the original basis, i.e., the shift of the unperturbed Coulomb wave. Thus using for each α subspace a basis built by taking products of the relevant ion state times the orbitals obtained by diagonalizing the hydrogenic Hamiltonian on the appropriate l_{α}, m_{α} GTO set, Eq. (9) may be applied to recover the total phase shift of the resulting pwc states. The justification of this lies in the assumption that the difference between the effective field due to the charge distribution of the ion and a Coulomb field is non-negligible only in the limited region of space including the charge of the ion, where the GTOs may satisfactorily reproduce the wave functions.

This technique was previously¹² labeled separate channel K matrix (SC-KM), as opposed to the conventional standard many-channel K matrix, hereafter indicated as interacting channel K matrix (IC-KM).

Notice that the procedure sketched above applies as an alternative approach to the interpolation technique normally employed to compute phase shifts with the oscillating STOCOS functions as well as in cases where the L^2 representation of the pwc basis functions does not display the correct long-range behavior.

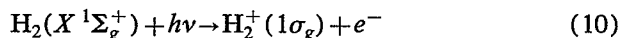
A SC-KM calculation thus yields the appropriate phase shifts of the pwc basis functions $|\Phi_{\alpha,E}\rangle$. The subsequent standard IC-KM step furnishes the continuum wave functions and the transition density matrices.

The technique presented in this section is easily generalized and adapted to the random phase approximation (RPA), thus allowing one to extract from the results of an L^2 finite orbital-basis calculation the matrix elements for transition to the degenerate continua. The corresponding equations can be found in the literature.^{10,15}

III. CALCULATIONS

A. Generalities

As a test of the efficiency of the method outlined in the preceding section for molecules, the photoionization process



was studied.

Although the system is very simple, it provides a stringent test of the method and, in spite of the vast literature on the asymmetry parameter, there appears to still be some disagreement between experiment and theory.^{5,16,17}

All our pwc basis functions were determined for each (l,m) by solving the eigenvalue equation for the appropriate singlet spin static-exchange (SEA) Hamiltonian operator upon the L^2 basis. To compute their phase shifts, SC-KM calculations were performed on the same GTO sets used for the IC-KM calculations. As the basis for the purely Coulomb Hamiltonian, 20–25 even tempered GTO functions per (l,m) , centered on the midpoint of the internuclear axis of the molecule, were employed. The resulting GTO approximation of pure Coulomb waves is of course satisfactory only for a short spatial range (up to about

15–20 a.u. for the lowest lying continuum orbitals) and is poorly suited to represent the oscillatory behavior of the high-energy waves, although it appears to correctly mimic lower-energy waves.

In this case, as already indicated and as in the case of the helium atom¹² the inadequacy of the GTOs, together with the relative complexity of the potential produced by the parent ion, prevent one from obtaining both the phase shifts and the energy normalization of the pwc states via an analysis of their long-range features.

The energy normalization constants for the pwc states were estimated from the energy spacings. If $|\varphi_{\beta j}\rangle$ is the L^2 approximation for the pwc wave function $|\Phi_{\beta E}\rangle$, then an energy-normalized wave packet is obtained as $|\Phi_{\beta j}\rangle = C_{\beta j} |\varphi_{\beta j}\rangle$, where $C_{\beta j}^2 \approx 2/(E_{\beta, j+1} - E_{\beta, j-1})$.

These estimates prove in general to be surprisingly accurate,¹⁸ especially for energies not too close to the ionization limit, where the continuum wave packets may mix strongly with the higher lying Rydberg states. An alternative approach, consisting of fitting with a polynomial function the histogram representing the density of states $n(\epsilon)$ (ϵ being the photoelectron energy) and dividing the continuum wave functions by the square root of the inverse of the derivative $dn(\epsilon)/d\epsilon$, lead to negligible variations only in the critical region near the ionization threshold.

The results, obtained using GTO basis sets, were compared to those given in the one center approximation (OCE) by large sets including STO, hydrogenic, and STOCOS functions.¹³ These last functions proved to be extremely effective in calculations on molecular continua,¹³ and it should be remembered that the OCE is expected to be a valid approach to the description of the ground and excited states of the hydrogen molecule.

The techniques employed to project the K -matrix equations (4) and (5) on a discretized set and to solve them are described extensively elsewhere.^{10,18}

Three different approximations were adopted—a SEA approach, the RPA, and a CI approximation for both the ground and final continuum electronic states.

The SEA and RPA calculations can account only for the background continuum and for shape resonances when present. All other resonance structures of the Feshbach type may be accounted for only in the CI calculations. The range of photon energies where the SEA and RPA calculations yield valid results is thus limited to values low enough so as not to enter the region where Feshbach resonances might appear (approximately up to 30 eV).

To study the effect of correlation on the properties investigated, calculations were performed in which both the ground and final states of the molecule were described by suitable CI expansions. In this case, several localized states were included, together with the appropriate pwc channels, in the CI expansions for the final states. This allowed the extension of the range of photon energies to the region around and beyond 30 eV, where resonance structures result from the interaction of the ionization continuum with doubly excited states.^{16,19,20}

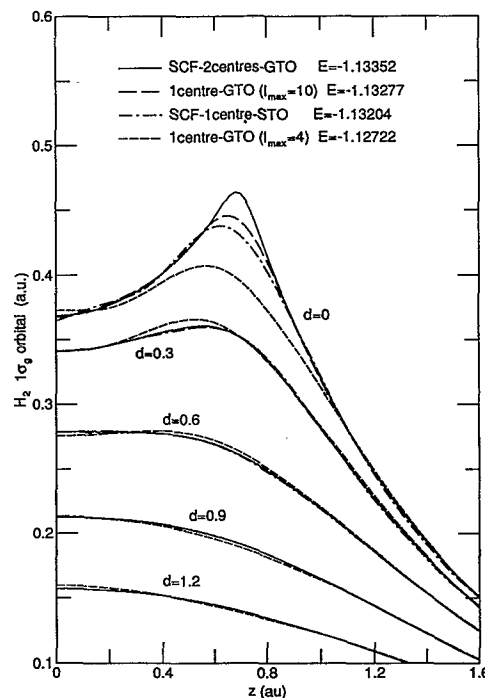


FIG. 1. A plot of the $1\sigma_g$ ground state occupied orbital of H₂ as a function of the internuclear coordinate z for some fixed values d of the $x(y)$ coordinate. Four different expansion sets (see the text for more details).

B. Results and discussion

An equilibrium distance of 0.7408 Å was assumed.

A two-center set of uncontracted Cartesian GTO functions ($\{10s3p1d\}$) was employed to describe the self-consistent field (SCF) ground state of the hydrogen molecule. The geometrical set employed for the $l=0$ symmetry had exponents determined by the rule $\alpha_j = \alpha_1 C^{j-1}$, $j=1, 10$ with $\alpha_1=0.028\,951\,4$ and $C=2.587\,417\,1$. Subsets of the exponents used for the s symmetry were chosen for the p (α_j , $j=4, 6$) and d (α_5) symmetries. This basis gives a SCF ground state energy which is only 0.0001 a.u. higher than the best value obtained in the highly accurate calculations by Wells and Wilson.²¹ The $1\sigma_g$ occupied orbital was then projected on a large GTO set centered on the center of the molecule. Geometrical sets including functions with l quantum number up to 10 (a total of 31 GTOs) were employed to project more than 99.999% of both the occupied $1\sigma_g$ and antibonding $1\sigma_u$ orbitals, i.e.,

$$|\langle 1\sigma_g(\text{TC}) | 1\sigma_g(\text{CM}) \rangle|^2 \geq 0.999\,998, \quad (11)$$

$$|\langle 1\sigma_u(\text{TC}) | 1\sigma_u(\text{CM}) \rangle|^2 \geq 0.999\,998, \quad (12)$$

where TC indicates “expanded on a two-center set,” while CM indicates “projected on the nuclear center of the molecule.” Notice that the orbital labeled $1\sigma_u(\text{TC})$ in Eq. (12) was obtained by solving the appropriate two-center SEA equation.

Figure 1 shows the behavior of the $1\sigma_g$ orbital of the hydrogen molecule in its ground state resulting from (i) the original standard two-center expansion;²² (ii) the projection onto the one-center GTO sets with maximum l

quantum number of 4; (iii) the projection onto the one-center GTO sets with maximum l quantum number of 10; and (iv) an OCE calculation.¹³ The nuclei are along the z axis. The amplitude of the $1\sigma_g$ orbital as a function of z for a few fixed values d of the $x(y)$ coordinate are plotted. As would be expected, the four functions differ in the region of space where they are expected to mimic the nuclear cusp. Notice that s and d functions alone are sufficient to project more than 99.99% of the occupied $1\sigma_g$ orbital and that the high- l quantum number functions in the projection basis are in practice needed only to improve the description of the nuclear cusp region, which on the other hand also means recovering up to 99.97% of the ground state two-center SCF energy. In fact, while the two-center Hartree-Fock energy with our 50 GTO basis described above was $-1.133\,520$ a.u. ($1\sigma_g$ orbital energy $\epsilon_{1\sigma_g} = -0.594\,66$ a.u.), the mean value of the total energy computed using the $1\sigma_g$ orbital of the hydrogen molecule projected on the nuclear center of mass upon the high- l GTO basis was $-1.132\,773$ a.u. and the corresponding $1\sigma_g$ energy was $-0.594\,32$ a.u. It should be added, however, that the effect of high- l quantum number functions in the projection of the occupied orbital upon the continuum properties in the case of H₂ appears to be completely negligible. Notice also that a rather limited 27 STO ($\langle 5s4d1g1i \rangle$) OCE calculation gives an energy of $-1.132\,039$ a.u. ($\epsilon_{1\sigma_g} = -0.594\,12$ a.u.).

The projection, though not strictly needed in this case, was used to give a complete test of the whole technique, which will be applied in the future to larger and more challenging diatomic molecules. In such cases, projecting the occupied orbitals (needed in the construction of the effective Hamiltonian) on a one-center basis will probably be expedient, especially when high angular quantum number partial waves play an important role, whereas accurate multicenter two-electron integral routines involving these high- l waves do not appear to be available.

Since the process under study leaves the H₂⁺ ion in its ground state, the waves involved are those belonging to the σ_u and π_u representation of the $D_{\infty h}$ point group. With the origin on the center of the molecule and the nuclei lying on the z axis, partial waves with $l=2L+1$, $m=0$, $L=0, \dots, \infty$ contribute to the σ_u channel, while partial waves with $l=2L+1$, $m=\pm 1$, $L=0, \dots, \infty$ contribute to the π_u ionization channel.

Figure 2 shows the phase shifts computed as discussed above for the (0,0), (1,0), and (1,±1) pwc states in the frozen orbital approximation, i.e., with the $1\sigma_g$ orbital arising from a SCF calculation on the neutral molecule. The results are compared with those obtained in the OCE approximation using large sets of STO and STOCOS functions (70–80 on the average) by analyzing the long-range behavior of the orbitals lying in the continuum. The first partial wave, belonging to the σ_g symmetry, although not involved in the actual photoionization process, exhibits by far the largest phase shift, at least an order of magnitude larger than that of the (1,0) and (1,±1) partial waves—it is consequently the most suitable for comparison purposes and to analyze the efficiency of the technique. Notice that

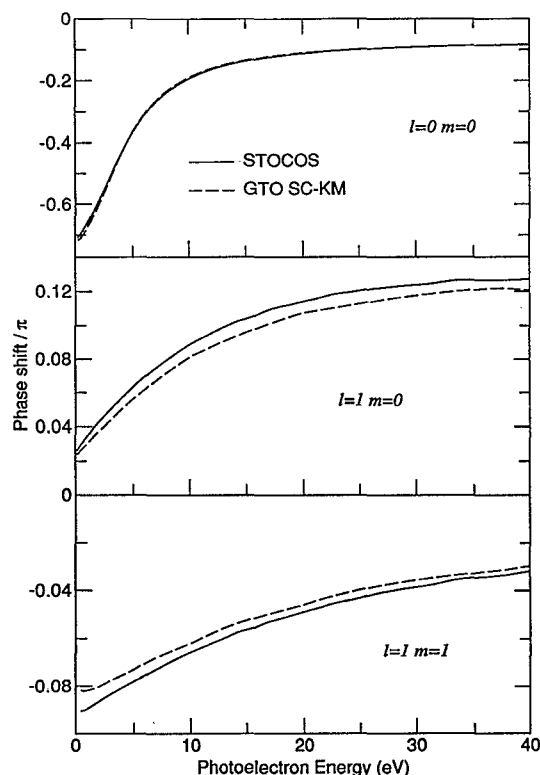


FIG. 2. Phase shifts of the (0,0), (1,1), and (1,−1) partial waves in units of π as a function of the photoelectron energy in electron volts. Standard KM calculation on a STOCOS expansion set (continuous line). SC-KM calculation upon a GTO expansion set (broken line).

the phase shifts of the higher- l partial waves are even smaller than those of the p waves. Hence the technique based on the use of Eq. (9) works remarkably well with short-range basis functions as the GTOs, and furnishes phase shifts which are easily comparable to those yielded by more suitable “continuum”-devised functions. From the data of Fig. 2, it appears that the largest difference between the two curves for each of the three partial waves does not exceed 0.01 rad/ π .

Figures 3–6 summarize the results obtained for both the integral photoionization cross section $\sigma^{(1)}(\omega)$ and the asymmetry parameter $\beta(\omega)$ defined by the relation

$$\frac{d\sigma^{(1)}(\omega)}{d\Omega_k} = \frac{\sigma^{(1)}(\omega)}{4\pi} [1 + \beta(\omega)P_2(\cos \theta_k)] \quad (13)$$

as a function of the photon energy ω . In Eq. (13), Ω_k is the solid angle (θ_k, ϕ_k) along the direction $\mathbf{k}$ of the photoelectron and P_j indicates the Legendre polynomial of order j .

The figures show the results obtained for both the length gauge (LG) and velocity gauge (VG) choices of the form of the interaction operator in the different approximations employed. Partial waves with $l=1$ (23 basis functions), $l=3$ (21 functions), and $l=5$ (18 functions) were employed in all calculations. Waves with $l=7$ proved to have a completely negligible effect on the properties shown. Several available experimental values, with their associated uncertainties when given, are also reported in the figures.

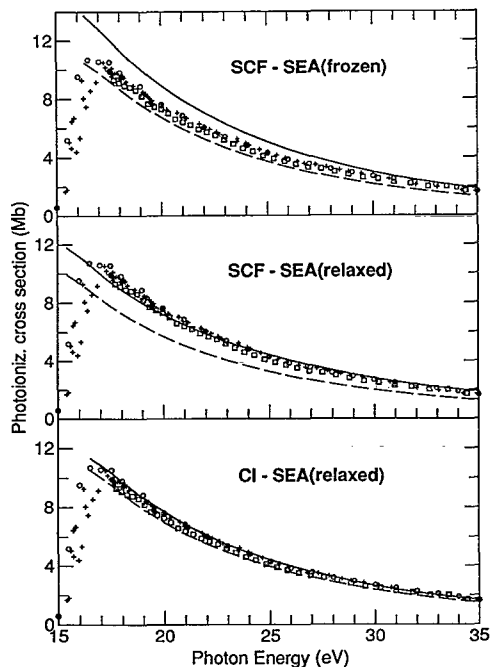


FIG. 3. Integral cross section (Mb) as a function of the photon energy in electron volts for the photoionization process starting from the ground state of H_2 . Three different combinations of approximations for the initial (left in the label) and final (right in the label) states. SCF-SEA (frozen) (top); SCF-SEA (relaxed) (mid); CI-SEA (relaxed) (bottom) (see also the text). Length gauge results (continuous lines); velocity gauge results (broken lines). Experimental values: $\circ$ —Backx *et al.* (Ref. 24); $+$ —Samson (Ref. 25); $\square$ —Lee *et al.* (Ref. 26).

The photoionization cross section obtained in three different SEA calculations, labeled SCF-SEA (frozen), SCF-SEA (relaxed), and CI-SEA (relaxed), respectively, is reported in Fig. 3. The labels identify the approximations employed for the ground and the final states, the former being described either by a SCF or a CI expansion. The configurations included in the CI expansion of the ground $1\Sigma_g^+$ state were selected perturbatively from all those obtainable with two electrons in orbitals of the type $\sigma_g, \sigma_u, \pi_g, \pi_u, \delta_g,$ and δ_u (over 150 configuration state functions). The total energy was of $-1.172\,489$ a.u., i.e., $0.001\,99$ a.u. above the exact variational limit.²³ In the SEA (frozen) approach, the pwc states were obtained by diagonalizing an effective Hamiltonian including the potential arising from the ground state SCF $1\sigma_g$ occupied orbital of the neutral molecule. In the SEA (relaxed) instead the partial waves were determined in the potential generated by the $1\sigma_g$ orbital of the H_2^+ ion.

The results for the asymmetry parameter $\beta(\omega)$ practically coincide in all three cases with those obtained in the RPA (Fig. 4) and were not reported in Fig. 3.

The agreement between the LG and VG results in Fig. 3 increases steadily in moving from the SCF-SEA (frozen) through the SCF-SEA (relaxed) to the CI-SEA (relaxed), the two curves converging to the experimental values^{24–26} in the 17–35 eV photon energy range of the figure. Notice, however, that the experiment misses the structure due to

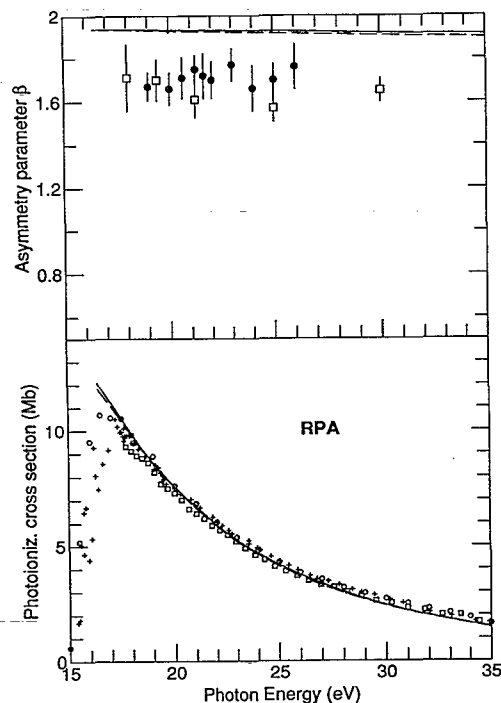


FIG. 4. RPA results for the integral cross section (Mb) (bottom) and asymmetry parameter β (top) as a function of the photon energy in electron volts for the photoionization process starting from the ground state of H_2 . Length gauge results (continuous lines); velocity gauge results (broken lines). Experimental values for the integral cross section: $\circ$ —Backx *et al.* (Ref. 24); $+$ —Samson (Ref. 25); $\square$ —Lee *et al.* (Ref. 26). Experimental values for the asymmetry parameter: $\bullet$ —Southworth *et al.* (Ref. 30) $\square$ —Marr *et al.* (Ref. 31).

the onset of Feshbach type resonances at photon energies beyond 30 eV.

The agreement between LG and VG results becomes almost complete for the curves of Fig. 4, where both the integral cross section and the $\beta(\omega)$ asymmetry parameter computed in the RPA are shown. The equivalence of the two gauges, an intrinsic property of a RPA calculation performed on a complete basis set, is fulfilled in this special case in a surprising extent due to both a satisfying choice of the basis and the peculiarity of the two electron system under study. Our results for the integral cross section fit remarkably well the experimental points for photon energies beyond 17 eV.

A CI treatment of both the ground and final continuum states, beside furnishing results for the integral cross section in good agreement with experiment, introduces the resonant structure around and beyond 30 eV shown in Fig. 5. A detailed view of the 30–35 eV photon energy range is given in Fig. 6, where the contributions of the channels leading to final $1\Pi_u$ and $1\Sigma_u^+$ states to the integral cross sections are plotted separately.

Localized states corresponding to configurations of the type $\sigma_u\sigma_g, \pi_u\pi_g, \delta_u\delta_g, \sigma_g\pi_u, \sigma_u\pi_g, \delta_u\pi_g,$ and $\delta_g\pi_u$ entered the CI expansions for the final $1\Sigma_u^+$ and $1\Pi_u$ states together with the appropriate $1\sigma_g\sigma_u$ and $1\sigma_g\pi_u$ configurations including the background continuum pwc's.

The overall structure, leading to peaks in the integral

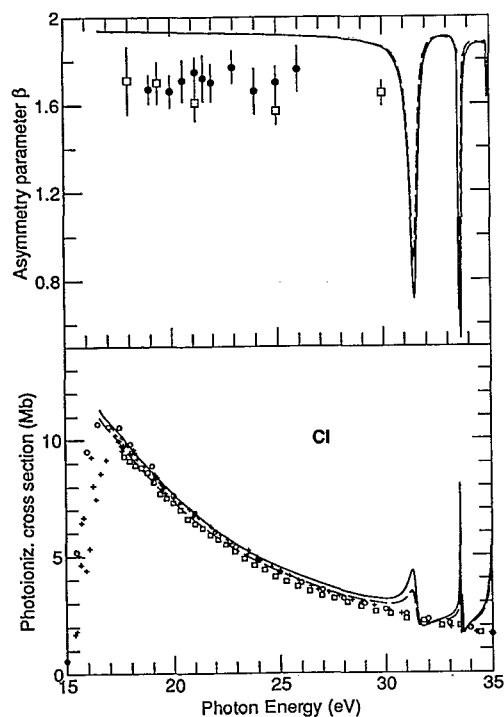


FIG. 5. CI results for the integral cross section (Mb) (bottom) and asymmetry parameter (top) as a function of the photon energy in electron volts for the photoionization process starting from the ground state of H₂ (see the text). Length gauge results (continuous lines); velocity gauge results (broken lines). Experimental values for the integral cross section: ○—Backx *et al.* (Ref. 24); +—Samson (Ref. 25); □—Lee *et al.* (Ref. 26). Experimental values for the asymmetry parameter: ●—Southworth *et al.* (Ref. 30); □—Marr *et al.* (Ref. 31).

cross section and troughs in the asymmetry parameter in Figs. 5 and 6, results from the interaction of the background continuum with a $1\sigma_u 1\pi_g$ doubly excited state of $p \rightarrow d$ character in the $^1\Pi_u$ manifold and a family of $1\sigma_u n\sigma_g$ double excitations of mixed $p \rightarrow s$ and $p \rightarrow d$ character in the $^1\Sigma_u^+$ manifold.

There has been no experimental evidence to date of the resonant structure shown in Figs. 5 and 6. On the other hand, it was the subject of quite detailed theoretical studies by others.^{20,27} Our results show an overall agreement with these previous theoretical estimates. The autoionizing states appear to be narrow, the photoelectron ejection becoming thus almost isotropic, i.e., $\beta(\omega)$ deepening down toward 0. The $1\sigma_u(2p)1\pi_g(3d)$ resonance around 33.6 eV appears to be very narrow compared to the others, as was found by Raseev.²⁰

All in all, however, all our results for the integral cross section agree well with other previous accurate theoretical values.⁴

As far as the asymmetry parameter is concerned, noticeable discrepancies between experiment and theory remain. While our values for $\beta(\omega)$ vary only by 1%–2% in the range of photon frequencies not including resonant structures, and are in excellent overall agreement with other previous calculations,^{5,27–29} experiment predicts much lower averages^{30,31} to which our calculated values should be compared. Even taking into account the experi-

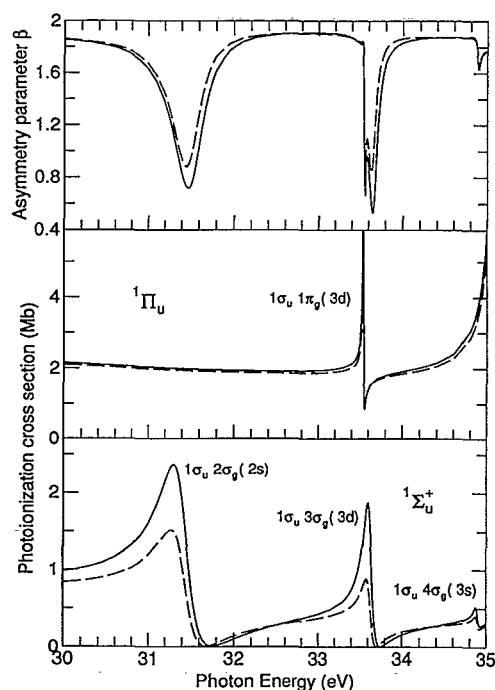


FIG. 6. Partial cross section (Mb) for the $^1\Sigma_u^+$ (bottom) and the $^1\Pi_u$ (mid) channels and asymmetry parameter (top) as a function of the photon energy in electron volts in the 30–35 eV range for the photoionization process starting from the ground state of H₂. CI results (see the text). Length gauge results (continuous lines); velocity gauge results (broken lines). The dominant configuration for each autoionizing state is indicated.

mental error bars and our estimated error bars, which should not modify the curves by more than 1%–2%, these differences cannot be reconciled.

One could note that simple arguments, based on the strong analogies existing between the hydrogen molecule and its closest atomic isoelectronic system (the helium atom) would suggest an asymmetry parameter for H₂ very close to the limiting value of 2. In fact, the occupied $1\sigma_g$ in the ground state of H₂ has an overwhelming dominance of s character, while the contribution of p partial waves largely overshadows that of the other symmetry types.

On the other hand, it is difficult to believe that for this relatively simple system, using several different and increasingly sophisticated approaches, one could obtain rather good estimates of the total cross section, but values of the asymmetry parameters 15% to 20% in error. In order to ascertain if a better agreement might be obtained by including vibrational interaction, calculations of $\beta(\omega)$ were carried out also at internuclear distances of 0.6615 and 0.8202 Å. Results for both the total and the differential cross section were found to be almost identical to those at the reference equilibrium geometry. Although this provides only a rough test of a possible vibrational effect, the stability of the results obtained at all three different internuclear distances suggests that the effect of vibrations, at least in the Born–Oppenheimer approximation, is insufficient to explain the discrepancy. The validity of the rotational spherical average approximation for a molecule of small moment of inertia might also be questioned; but

again this seems a quite unlikely explanation of the disagreement.

IV. CONCLUSIONS

A proper use of GTO basis sets has yielded quite accurate integral and differential cross sections of molecular hydrogen. The importance of these results goes well beyond the actual system under study. In principle, it should be possible to adapt some of the quite sophisticated codes available for molecular structure calculations to include the procedures discussed in this work. This would permit the accurate study of properties of the electronic continuum for a wide range of systems.

The extension of the techniques to larger molecules will not of course be completely straightforward. The approximations introduced by the projection of the reference state occupied orbitals onto the center of expansion of the continuum partial waves, although completely negligible for the H₂ molecule, need to be overcome in more challenging molecular systems. The first priority will thus be to adapt a multicenter GTO integral generation package, which should be able to compute two-electron integrals between GTOs having in principle quite high l quantum numbers, to the static-exchange approximation. For larger molecular systems, there is no guarantee that the pwc expansion can be easily truncated at low values of l , and much larger GTO sets will probably be needed. As the number of interacting channels increases, so will the dimensions of the calculation. Work is in progress in this group to tackle these problems and to move to larger linear molecular systems.

On the other hand, the hydrogen molecule itself apparently still deserves further attention. The intriguing discrepancies existing between theory and experiment in the differential cross section will clearly not be resolved without renewed efforts on all sides. As far as our group is concerned, we plan to continue the study of H₂ by focusing on approximations which will allow us to estimate the effect of vibrations on the properties of the electronic continuum. Shapes and position of the autoionization resonances will also be studied in more detail.

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